

WEEK 3 SOC INCIDENT REPORT

Incident Title: Malware Infection & C2 Beaconing from Internal Host

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Severity: High

A. Executive Summary

A workstation within the **10.1.17.0/24** LAN segment initiated suspicious DNS queries to a fake Google Authenticator domain and subsequently communicated with several high-risk external IPs associated with command-and-control (C2) activity. Analysis of the packet capture (pcap) confirms beaconing behavior, possible credential theft, and potential compromise of an Active Directory joined host.

Immediate containment is required to prevent further lateral movement, exfiltration, or domain compromise.

B. Indicators of Compromise (IOCs)

1. The IP address of the infected Windows client:

 **10.1.17.215** (infected host observed in pcap)

ip.src== 10.1.17.0/24 && ip.dst != 10.1.17.0/24						
No.	Time	Source	Destination	Protocol	Length	Info
75	4.310252	10.1.17.215	23.40.146.4	TCP	60	50085 → 443 [ACK] Seq=1 Ack=1 Win=65280 L
76	4.311084	10.1.17.215	23.40.146.4	TLSv1.2	271	Client Hello (SNI=kv801.prod.do.dsp.mp.mi
82	4.365798	10.1.17.215	23.40.146.4	TCP	60	50085 → 443 [ACK] Seq=218 Ack=2753 Win=65
83	4.365799	10.1.17.215	23.40.146.4	TCP	60	50085 → 443 [ACK] Seq=218 Ack=5473 Win=65
85	4.407389	10.1.17.215	23.40.146.4	TLSv1.2	180	Client Key Exchange, Change Cipher Spec,
87	4.442457	10.1.17.215	23.40.146.4	TLSv1.2	382	Application Data
89	4.492788	10.1.17.215	52.156.123.84	TCP	66	50086 → 443 [SYN] Seq=0 Win=65535 Len=0 M
90	4.528764	10.1.17.215	23.40.146.4	TCP	60	50085 → 443 [ACK] Seq=672 Ack=7628 Win=65
92	4.585492	10.1.17.215	52.156.123.84	TCP	60	50086 → 443 [ACK] Seq=1 Ack=1 Win=65280 L
93	4.586371	10.1.17.215	52.156.123.84	TLSv1.2	269	Client Hello (SNI=geo.prod.do.dsp.mp.micr
96	4.671631	10.1.17.215	52.156.123.84	TCP	60	50086 → 443 [ACK] Seq=216 Ack=2432 Win=65
97	4.674112	10.1.17.215	52.156.123.84	TLSv1.2	212	Client Key Exchange, Change Cipher Spec,
102	4.764541	10.1.17.215	52.156.123.84	TCP	60	50086 → 443 [ACK] Seq=374 Ack=2552 Win=65
103	4.765411	10.1.17.215	52.156.123.84	TLSv1.2	141	Application Data
104	4.765411	10.1.17.215	52.156.123.84	TLSv1.2	229	Application Data
105	4.765411	10.1.17.215	52.156.123.84	TLSv1.2	92	Application Data
108	4.833964	10.1.17.215	23.220.102.9	TCP	66	50087 → 80 [SYN] Seq=0 Win=65535 Len=0 MS
110	4.880798	10.1.17.215	23.220.102.9	TCP	60	50087 → 80 [ACK] Seq=1 Ack=1 Win=65280 Le
111	4.880969	10.1.17.215	23.220.102.9	HTTP	165	GET /connecttest.txt HTTP/1.1
115	4.899916	10.1.17.215	52.156.123.84	TCP	60	50086 → 443 [ACK] Seq=674 Ack=3167 Win=64
116	4.900250	10.1.17.215	52.156.123.84	TCP	60	50086 → 443 [FIN, ACK] Seq=674 Ack=3167 W
120	4.930864	10.1.17.215	23.220.102.9	TCP	60	50087 → 80 [ACK] Seq=112 Ack=189 Win=6528
121	4.930864	10.1.17.215	23.220.102.9	TCP	60	50087 → 80 [FIN, ACK] Seq=112 Ack=189 Win
125	4.990905	10.1.17.215	52.156.123.84	TCP	60	50086 → 443 [RST, ACK] Seq=675 Ack=3209 W
126	4.990905	10.1.17.215	52.156.123.84	TCP	60	50086 → 443 [RST] Seq=675 Win=0 Len=0
131	5.511793	10.1.17.215	20.10.31.115	TCP	66	50088 → 443 [SYN] Seq=0 Win=65535 Len=0 M
133	5.600487	10.1.17.215	20.10.31.115	TCP	60	50088 → 443 [ACK] Seq=1 Ack=1 Win=65280 L
134	5.601537	10.1.17.215	20.10.31.115	TLSv1.2	232	Client Hello (SNI=client.wns.windows.com)
140	5.700813	10.1.17.215	20.10.31.115	TCP	60	50088 → 443 [ACK] Seq=179 Ack=2921 Win=65
141	5.702778	10.1.17.215	20.10.31.115	TLSv1.2	212	Client Key Exchange, Change Cipher Spec,

2. The MAC address of the infected Windows client:

➡ 00:d0:b7:26:4a:74

The image shows a Wireshark packet capture window titled 'arp'. The packet list on the left shows four packets, with packet 169 selected. The packet details pane on the right shows the structure of the selected packet, which is an ARP response. The packet is an Ethernet II frame with source Intel_26:4a:74 (00:d0:b7:26:4a:74) and destination Dell_7f:09:5d (00:24:e8:7f:09:5d). The ARP section shows the sender's MAC address as Intel_26:4a:74 (00:d0:b7:26:4a:74) and the target's IP address as 10.1.17.2.

No.	Time	Source	Destination	Protocol	Length	Info
163	7.675442	Intel_26:4a:74	Broadcast	ARP	60	Who has 10.1.17.2? Tell 10.1.17.215
164	7.675443	Dell_7f:09:5d	Intel_26:4a:74	ARP	60	10.1.17.2 is at 00:24:e8:7f:09:5d
168	7.893453	Dell_7f:09:5d	Intel_26:4a:74	ARP	60	Who has 10.1.17.215? Tell 10.1.17.2
169	7.893987	Intel_26:4a:74	Dell_7f:09:5d	ARP	60	10.1.17.215 is at 00:d0:b7:26:4a:74

Frame 169: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
Ethernet II, Src: Intel_26:4a:74 (00:d0:b7:26:4a:74), Dst: Dell_7f:09:5d (00:24:e8:7f:09:5d)
Destination: Dell_7f:09:5d (00:24:e8:7f:09:5d)
Source: Intel_26:4a:74 (00:d0:b7:26:4a:74)
Type: ARP (0x0806)
[Stream index: 2]
Padding: 00000000000000000000000000000000
Address Resolution Protocol (reply)
Hardware type: Ethernet (1)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: reply (2)
Sender MAC address: Intel_26:4a:74 (00:d0:b7:26:4a:74)
Sender IP address: 10.1.17.215
Target MAC address: Dell_7f:09:5d (00:24:e8:7f:09:5d)
Target IP address: 10.1.17.2

(Extracted from ARP response: “10.1.17.215 is at 00:d0:b7:26:4a:74”)

3. The host name of the infected Windows client:

➡ DESKTOP-L8C5GSJ (Hostname)

The image shows a Wireshark packet capture window titled 'nbns'. The packet list on the left shows a series of NetBIOS Name Service registration packets from 10.1.17.215 to 10.1.17.255. The packet details pane on the right shows the structure of the selected packet, which is a NetBIOS Name Service registration packet. The packet is an Ethernet II frame with source Intel_26:4a:74 (00:d0:b7:26:4a:74) and destination Broadcast (ff:ff:ff:ff:ff:ff). The NetBIOS Name Service section shows the transaction ID as 0xd632, flags as 0x2910, and opcode as Registration. The query section shows the name as DESKTOP-L8C5GSJ (Workstation/Redirector), type as NB (32), and class as IN (1).

No.	Time	Source	Destination	Protocol	Length	Info
19	0.878719	10.1.17.215	10.1.17.255	NBNS	110	Registration NB DESKTOP-L8C5GSJ<00>
20	0.878719	10.1.17.215	10.1.17.255	NBNS	110	Registration NB DESKTOP-L8C5GSJ<20>
21	0.878896	10.1.17.215	10.1.17.255	NBNS	110	Registration NB BLUEMOONTUESDAY<00>
42	0.855475	10.1.17.215	10.1.17.255	NBNS	110	Registration NB BLUEMOONTUESDAY<00>
43	0.855476	10.1.17.215	10.1.17.255	NBNS	110	Registration NB DESKTOP-L8C5GSJ<20>
44	0.855612	10.1.17.215	10.1.17.255	NBNS	110	Registration NB DESKTOP-L8C5GSJ<00>
47	1.613913	10.1.17.215	10.1.17.255	NBNS	110	Registration NB DESKTOP-L8C5GSJ<00>
48	1.613913	10.1.17.215	10.1.17.255	NBNS	110	Registration NB DESKTOP-L8C5GSJ<20>
49	1.614070	10.1.17.215	10.1.17.255	NBNS	110	Registration NB BLUEMOONTUESDAY<00>
51	2.372121	10.1.17.215	10.1.17.255	NBNS	110	Registration NB BLUEMOONTUESDAY<00>
52	2.372121	10.1.17.215	10.1.17.255	NBNS	110	Registration NB DESKTOP-L8C5GSJ<20>
53	2.372279	10.1.17.215	10.1.17.255	NBNS	110	Registration NB DESKTOP-L8C5GSJ<00>

Frame 19: Packet, 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface 0
Ethernet II, Src: Intel_26:4a:74 (00:d0:b7:26:4a:74), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 10.1.17.215, Dst: 10.1.17.255
User Datagram Protocol, Src Port: 137, Dst Port: 137
NetBIOS Name Service
Transaction ID: 0xd632
Flags: 0x2910, Opcode: Registration, Recursion desired, Broadcast
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
DESKTOP-L8C5GSJ<00>: type NB, class IN
Name: DESKTOP-L8C5GSJ (Workstation/Redirector)
Type: NB (32)
Class: IN (1)
Additional records
DESKTOP-L8C5GSJ<00>: type NB, class IN

(Observed from NBNS/NetBIOS broadcast traffic)

4. The user account name from the infected Windows client:

→ KRB5-NT-PRINCIPAL (1)

The image shows a Wireshark capture of Kerberos traffic. The packet list on the left shows a series of Kerberos messages. The packet details pane on the right shows the structure of a TGS-REP message. The 'cname' field is expanded, and the 'name-type' is highlighted as 'KRB5-NT-PRINCIPAL (1)'. The packet bytes pane on the right shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
30	0.128268	10.1.17.215	10.1.17.2	LDAP	748	bindRequest(11) "<ROOT>" sasl
32	0.129934	10.1.17.2	10.1.17.215	LDAP	264	bindResponse(11) success
250	14.368083	10.1.17.215	10.1.17.2	KRB5	288	AS-REQ
251	14.369177	10.1.17.2	10.1.17.215	KRB5	262	KRB Error: KRB5KDC_ERR_PREAUTH_REQUIRED
258	14.374722	10.1.17.215	10.1.17.2	KRB5	368	AS-REQ
260	14.376723	10.1.17.2	10.1.17.215	KRB5	399	AS-REP
269	14.378127	10.1.17.215	10.1.17.2	KRB5	307	TGS-REQ
272	14.380720	10.1.17.2	10.1.17.215	KRB5	329	TGS-REP
293	14.528203	10.1.17.215	10.1.17.2	KRB5	483	TGS-REQ
296	14.528453	10.1.17.2	10.1.17.215	KRB5	401	TGS-REP
302	14.530188	10.1.17.215	10.1.17.2	DCERPC	851	Bind: call_id: 2, Fragment: Single, 3 con
304	14.531000	10.1.17.2	10.1.17.215	DCERPC	338	Bind_ack: call_id: 2, Fragment: Single, m
305	14.531217	10.1.17.215	10.1.17.2	DCERPC	274	Alter_context: call_id: 2, Fragment: Sing
319	14.537320	10.1.17.215	10.1.17.2	LDAP	736	bindRequest(15) "<ROOT>" sasl
321	14.538167	10.1.17.2	10.1.17.215	LDAP	264	bindResponse(15) success
328	14.546374	10.1.17.215	10.1.17.2	LDAP	742	bindRequest(19) "<ROOT>" sasl
330	14.547144	10.1.17.2	10.1.17.215	LDAP	264	bindResponse(19) success
818	26.588158	10.1.17.215	10.1.17.2	DCERPC	851	Bind: call_id: 2, Fragment: Single, 3 con

Packet details for TGS-REP (296):

- Record Mark: 1863 bytes
- pvno: 5
- msg-type: krb-tgs-rep (13)
- crealm: BLUEMOONTUESDAY.COM
- cname:
 - name-type: KRB5-NT-PRINCIPAL (1)
 - cname-string: 1 item
- ticket
- enc-part
- Response to: 293
- Time from request: 1.251000 milliseconds
- Missing keytype 18 usage 8 or 9 missing in frame 296 key
- Expert Info (Warning/Decryption): Missing keytype 18
- Missing keytype 18 usage 8 or 9 missing in frame 29
- Severity level: Warning
- Group: Decryption

✓ To find the user account name, you must check:

- Kerberos (KRB5) > TGS-REQ > cname
- Domain Controller logs (Event ID 4624 / 4768 / 4769)
- EDR logs (if available)

5. The likely domain name for the fake Google Authenticator page:

→ google-authenticator.burleson-appliance.net (fake domain name of Google Authenticator landing page)

→ authenticatoor.org (observed SNI during TLS to an external IP)

The image shows a Wireshark packet capture of a DNS transaction. The packet list on the left shows a standard query (2322) and a standard query response (2323) for google-authenticator.burleson-appliance.net. The packet details pane on the right shows the response structure, including flags, questions, and answers. The answer section shows the query result: google-authenticator.burleson-appliance.net: type HTTPS, class IN, with a response time of 79.153800 milliseconds.

This comes from a DNS query made by the infected workstation.

6. The IP addresses used for C2 servers for this infection:

External IPs (observed C2 / suspicious) is gotten from the statistics conversation with more packets identified and the repeated TLS and HTTP beaconing in the pcap, the C2 infrastructure consists of:

- ➡ 45.125.66.32
- ➡ 45.125.66.252
- ➡ 82.221.136.26
- ➡ 5.252.153.241

Ethernet · 7	IPv4 · 144	IPv6	TCP · 421	UDP · 346	
Address A	Address B	Packets ^	Bytes	Stream	
10.1.17.215	45.125.66.32	10,940	10 MB		
10.1.17.215	5.252.153.241	9,076	7 MB		
10.1.17.215		4,359	1 MB		
10.1.17.215	82.221.136.26	2,470	2 MB		
10.1.17.215	45.125.66.252	1,369	107 kB		
10.1.17.215	23.55.125.176	1,018	708 kB		
10.1.17.215	204.79.197.203	594	261 kB		
10.1.17.215	199.232.214.172	556	514 kB		

These IPs show:

- Continuous beaconing
- Suspicious SNI (authenticatoor.org)
- Periodic HTTP GETs to nonsensical numeric paths
- TLS connections inconsistent with legitimate service

HTTP artifacts:

- Repeated GETs like /1517096937 to 5.252.153.241 returning 404 (beacon-like behavior).

Network traffic patterns:

- High-frequency TLS sessions between infected host and above IPs, small keepalive-style packets and application data (typical C2 beacons).

C. Key Facts Identified

Infected Endpoint

Attribute	Value
• Hostname	DESKTOP-L8C5GSJ
• IP Address	10.1.17.215
• MAC Address	00:d0:b7:26:4a:74
• Domain Membership	bluemoontuesday.com
• User Account	kRB5-NT-PRINCIPAL (1)

Malicious/Suspicious Domains

Domain	Purpose
• google-authenticator.burleson-appliance.net	Fake Google Authenticator phishing page
authenticatoor.org	Observed as SNI in malicious TLS session

Observed C2 / Malicious IP Addresses

IP Address	Description
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- **45.125.66.32** Primary, repeated TLS C2 beaconing
- **45.125.66.252** Secondary related infrastructure
- **82.221.136.26** TLS with suspicious SNI (*authenticatoor.org*)
- **5.252.153.241** HTTP beaconing, returning 404 endpoints typical of staging servers

D. Incident Timeline (from PCAP)

Time (approx)	Activity
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- T0 Host sends NBNS registrations announcing hostname **DESKTOP-L8C5GSJ**
- T1 DNS request for **google-authenticator.burleson-appliance.net** → indicates user interaction with malware/phishing
- T2 Host initiates TLS connections to **82.221.136.26** with SNI *authenticatoor.org*
- T3 Continuous outbound TLS sessions to **45.125.66.32** & **45.125.66.252** consistent with C2 beaconing
- T4 Repeated HTTP GET requests to **5.252.153.241** for unusual numeric paths (e.g., /1517096937) indicating malware heartbeat.

E. Impact / Risk

- Host compromise, likely initial user-driven (malicious landing page/ad → download).
- Potential for credential theft (user targeted by fake auth page), lateral movement (host talking to domain controller via Kerberos/LDAP seen in capture), and data exfiltration given persistent connections to external C2 IPs.
- Domain-joined Windows workstation in AD environment (domain bluemoontuesday.com) — risk of domain credential access and escalation.

F. Remediation

1. Isolate host 10.1.17.215 (network-level quarantine) — remove network connectivity immediately (disconnect from LAN or move to remediation VLAN).
2. Collect forensic images of the workstation (disk image, memory) before doing anything intrusive on the endpoint. Memory capture is especially important to recover malware process and possible in-memory credentials.
3. Preserve network evidence — secure the original pcap and logs; export additional PCAPs if needed for related hosts.
4. Identify other potentially impacted hosts — search for connections to the C2 IPs (45.125.66.32, 45.125.66.252, 82.221.136.26, 5.252.153.241) across firewall/proxy and endpoint telemetry.
5. Reimage the host and reintroduce to network only after full validation.
6. Search environment for other hosts contacting these IOCs and treat similarly.